Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI-Native Platforms · Agentic Systems · Human-in-the-Loop Design · LLM Evaluation · Python · LangGraph · FastAPI

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Virginia, USA — open to relocation

PROFESSIONAL SUMMARY

LeaseI is an 8-agent LangGraph AI platform I built: **Document Intelligence** that reasons over enterprise documents with citations and provenance tracking, and a **Support Investigation pipeline** that classifies tickets, retrieves business rules, configuration states, and historical incidents, then synthesizes ranked evidence-backed root-cause hypotheses — escalating to a human when confidence is too low. Human-in-the-loop is architectural: the LLM extracts and reasons; a human or deterministic gate makes the final call. Every answer is cited to its exact source clause or flagged unsupported — no silent failures. Eight years at Entrata: production data infrastructure at scale (1M+ records, AWS, PHI-compliant access-controlled pipelines, 100K MAU, 99.5% uptime). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

FEATURED PROJECTS

Leasel: AI Document Intelligence & Support Investigation Platform HITL · 8-AGENT

2026 · Open-source · Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a dual-mode internal developer platform for enterprise AI: **(1) Document Intelligence** — OCR-fallback ingestion → section-aware chunking → TF-IDF retrieval → confidence-scored answers with **100% citation correctness** on eval set and human-review gate for regulated decisions; **(2) Support Investigation** — 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional escalation edge when confidence falls below threshold.
- Enforced human-in-the-loop design throughout: LLM extracts and reasons; a human or deterministic logic makes the final eligibility/compliance/financial call — every answer includes provenance tracking (cited to exact source clause) or is explicitly flagged as unsupported and downgraded; validated end-to-end on misconfiguration root-cause scenario (income-eligible resident wrongly flagged → traced to misconfigured global setting).
- Achieved **93.6% precision** and **87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants); **pass@1 81.7%** and **pass@3 88.3%** via a custom evaluation pipeline and experiment tracking harness (pass@k, multi-trial sampling, regression gate, LLM-as-judge scoring) with feedback loops built in — every run reproducible and replayable.

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Systems · RAG · LangGraph · Compliance Automation

- Built an agentic mission-planning system as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) with a RAG layer over FAA Part 107 + operational-guidance corpus — every compliance check cited to source regulation; deterministic guardrails block non-compliant missions before export.

George Mason University

Aug 2024 – May 2026

Research Associate & Graduate Teaching Assistant — ML Pipelines · Anomaly Detection · Statistical Modeling

- Built multi-source prediction models (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied Isolation Forest to automate anomaly detection across 3 research pipelines; Tableau/ggplot2/Quarto dashboards drove **40% improvement** in faculty stakeholder comprehension.
- Mentored 600+ students across Python programming and database design (ER modeling, SQL, data validation).

Entrata — PropTech / Real Estate SaaS

Sep 2018 – Aug 2024

Senior Software Engineer — Data Infrastructure · AI Pipelines · Backend Systems · AWS · Access-Controlled Data

- Shipped 20+ production modules (Python, PHP, PostgreSQL, RabbitMQ, AWS) across **120+ Agile sprints** in a cross-functional environment (engineering, compliance, operations, product) — **1M+ property records**, **100K+ MAU**, **99.5% uptime** sustained 6+ consecutive months.
- Built access-controlled data infrastructure: Libsodium-encrypted PHI pipeline on AWS S3 — encrypted 100% of resident data (SSNs, phone numbers), updated 50%+ of codebase for compliance; **zero security incidents**.
- Shipped Entrata's first production AI pipeline on **1.6M financial records** (Python, PyTorch, OpenAI API) — automated data processing replacing manual Excel workflows; output validation gates and compliance controls at launch.
- Led microservices redesign (Docker): **50% scalability improvement**, **30% deployment time reduction**; optimized 200+ SQL queries across all client environments (**30% performance gain**).
- Built Python/Bash automation pipelines eliminating **80% of manual operational workload**; conducted 300+ peer code reviews; mentored 5+ junior engineers.

Entrata India (Associate / Trainee)

Jan 2017 – Sep 2018

Associate Software Engineer — Full-Stack · AWS S3 · Database Architecture

- Processed 1.6M HUD/Tax Credit income limit records; migrated thousands of client documents to AWS S3 (resolved 90% of legacy space constraints); redesigned document storage schema and database architecture, migrating hundreds of thousands of records to a modern data model with 6-month post-deployment stabilization.

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Tool-Use Patterns & Function Calling, Human-in-the-Loop (HITL), RAG Pipelines, Knowledge Retrieval, Embeddings, Semantic Search, ChromaDB, Vector Databases, LLM Evaluation, LLM-as-Judge, CI-Integrated Eval Pipelines, Provenance Tracking, Agent Observability, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI

Languages & Backend: Python, TypeScript, JavaScript, PHP, SQL, PL/SQL, R, Bash · REST APIs, Microservices, ETL, Data Pipelines, PostgreSQL, MySQL, SQLite, RabbitMQ

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Libsodium, Git, Linux

ML & Data: PyTorch, Scikit-learn, Isolation Forest, Random Forest, XGBoost, NLP, BERT, Hugging Face, spaCy, Feature Engineering, Tableau, Power BI, ggplot2, Jupyter

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University — 2013–2017